

PROBABILITY AND STATISTICS

A Simplified Textbook with Exact Mathematical Notation

Central tendency • positional measures • dispersion • skewness • moments • kurtosis • counting • conditional probability • total probability • Bayes' theorem

This book places the mathematical formula immediately beside the theory it represents. The notation follows the supplied lecture notes and worksheets, while the explanations are shortened for test and examination revision.

Basic Statistical Notation

Symbols used throughout the descriptive-statistics chapters

Suppose a dataset contains observations $x_1, x_2, x_3, \dots, x_n$. In a frequency distribution, each value x_i occurs f_i times.

Symbol	Meaning
x_i	Observation or class midpoint
f_i	Frequency of x_i
$N = \sum f_i$	Total number of observations
CF	Cumulative frequency
L	Lower limit of the required class
h	Class width
A	Assumed mean

Class midpoint

For continuous grouped data, every class is represented by its midpoint.

$$x_i = \frac{\text{Lower Limit} + \text{Upper Limit}}{2}$$

Continuous classes should have matching boundaries, such as 10-20, 20-30, 30-40, so that there are no gaps or overlaps.

Measures of Central Tendency

Mean, weighted mean, median, and mode

Arithmetic mean

The arithmetic mean is obtained by adding all observations and dividing by their number. It is the balancing point of the distribution and is most representative when the data are reasonably symmetric and contain no serious outliers.

$$\bar{X} = \frac{\sum_{i=1}^N X_i}{N}$$

$$\Sigma(X_i - \bar{X}) = 0$$

For discrete frequency data or continuous grouped data, every value is multiplied by its frequency.

$$\bar{X} = \frac{\sum f_i X_i}{\sum f_i}$$

Weighted mean

A weighted mean is used when observations have unequal importance. The weights may represent traffic, credit hours, quantities, or relative importance.

$$\bar{X}_W = \frac{\sum_{i=1}^n w_i X_i}{\sum_{i=1}^n w_i}$$

Assumed-mean method

A convenient central value A is chosen to reduce the size of the arithmetic. The deviations are measured from A.

$$d_i = X_i - A$$

$$\bar{X} = A + \frac{\sum d_i}{N}$$

$$\bar{X} = A + \frac{\sum f_i d_i}{\sum f_i}$$

Step-deviation method

When deviations are multiples of a common factor or class width h , the deviations are divided by h .

$$u_i = \frac{x_i - A}{h}$$

$$\bar{x} = A + \frac{\sum u_i}{N} \times h$$

$$\bar{x} = A + \frac{\sum f_i u_i}{\sum f_i} \times h$$

Correcting an incorrect mean

The reported mean is first converted back to a total. The incorrect observations are removed and the correct observations are inserted.

$$\Sigma x_{\text{wrong}} = N \bar{x}_{\text{wrong}}$$

$$\Sigma x_{\text{correct}} = N \bar{x}_{\text{wrong}} - \Sigma(\text{Wrong Values}) + \Sigma(\text{Correct Values})$$

$$\bar{x}_{\text{correct}} = \bar{x}_{\text{wrong}} + \frac{\Sigma(\text{Correct Value} - \text{Wrong Value})}{N}$$

Median

The median is the middle value after the observations have been arranged in ascending order. It is resistant to extreme values and is preferred for skewed distributions.

$$\text{Median Position} = \frac{N + 1}{2}$$

For even N , the two central observations are averaged.

$$\text{Median} = \frac{x_{(N/2)} + x_{(N/2+1)}}{2}$$

For discrete frequency data, calculate cumulative frequencies and select the value whose cumulative frequency first reaches or exceeds $(N+1)/2$.

For continuous grouped data, locate the class whose cumulative frequency first reaches $N/2$ and interpolate within that class.

$$\text{Median} = L + \frac{N/2 - CF_{<}}{f_m} \times h$$

Mode

The mode is the value occurring most frequently. For continuous grouped data, the class with the largest frequency is the modal class.

$$\text{Mode} = L + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$$

Here f_1 is the modal-class frequency, f_0 is the preceding frequency, and f_2 is the following frequency.

Mean, median, mode, and skewness

Mean = Median = Mode (Symmetric)

Mean > Median > Mode (Positive / Right Skew)

Mean < Median < Mode (Negative / Left Skew)

Positional Measures

Quartiles, deciles, percentiles, and the middle fifty percent

Quartiles

Quartiles divide an ordered dataset into four equal parts. $Q_1 = P_{25}$, $Q_2 = P_{50} = \text{Median}$, and $Q_3 = P_{75}$.

$$Q_k \text{ Position} = \frac{k(N + 1)}{4}$$

$$Q_k = L + \frac{kN/4 - CF_{<}}{fQ} \times h$$

Deciles

Deciles divide an ordered dataset into ten equal parts. $D_1 = P_{10}$, $D_5 = P_{50}$, and $D_9 = P_{90}$.

$$D_k \text{ Position} = \frac{k(N + 1)}{10}$$

$$D_k = L + \frac{kN/10 - CF_{<}}{fD} \times h$$

Percentiles

Percentiles divide an ordered dataset into one hundred equal parts.

$$P_k \text{ Position} = \frac{k(N + 1)}{100}$$

$$P_k = L + \frac{kN/100 - CF_{<}}{fP} \times h$$

$$\text{Bottom } x\% = P_x$$

$$\text{Top } x\% = P_{(100-x)}$$

$$\text{Top } 10\% = P_{90} \quad \text{Top } 5\% = P_{95} \quad \text{Top } 1\% = P_{99}$$

Interquartile range and quartile deviation

The interquartile range covers the middle fifty percent of the observations and is less affected by extreme values.

$$\text{IQR} = Q_3 - Q_1$$

$$\text{QD} = \frac{Q_3 - Q_1}{2}$$

$$\text{Coefficient of QD} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

Measures of Dispersion

Range, mean deviation, variance, standard deviation, and coefficient of variation

A central value describes where a distribution is located. Dispersion describes how far the observations spread around that central value.

Range

$$R = X_{\max} - X_{\min}$$

$$\text{Coefficient of Range} = \frac{X_{\max} - X_{\min}}{X_{\max} + X_{\min}}$$

Range is simple but depends only on the two extreme observations and is highly sensitive to outliers.

Mean deviation

Mean deviation measures the average absolute distance from the mean. Absolute values prevent positive and negative deviations from cancelling.

$$MD\bar{x} = \frac{\sum |x_i - \bar{x}|}{N}$$

$$MD\bar{x} = \frac{\sum f_i |x_i - \bar{x}|}{\sum f_i}$$

$$\text{Coefficient of MD} = \frac{MD\bar{x}}{\bar{x}}$$

Variance

Variance measures the average squared distance from the mean. Squaring gives greater importance to large deviations.

$$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$$

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$

$$\sigma^2 = \frac{\sum f_i (x_i - \bar{x})^2}{\sum f_i}$$

The computational forms are useful when sums of squares are supplied.

$$\sigma^2 = \frac{\sum x_i^2}{N} - \bar{x}^2$$

$$\sigma^2 = \frac{\sum f_i x_i^2}{N} - \bar{x}^2$$

Standard deviation

$$\sigma = \sqrt{\sigma^2}$$

$$s = \sqrt{s^2}$$

Variance has squared units. Standard deviation has the same units as the original observations. A smaller standard deviation indicates greater absolute consistency.

Coefficient of variation

$$CV = \frac{\sigma}{\bar{x}} \times 100\%$$

$$CV_a < CV_\beta \Rightarrow A \text{ is more consistent}$$

$$CV_a > CV_\beta \Rightarrow A \text{ has greater relative variation}$$

Coefficient of variation should be used when comparing datasets having different means, magnitudes, or units.

Shape of a Distribution

Skewness, central moments, and kurtosis

Karl Pearson's coefficient of skewness

$$Sk = \frac{\bar{x} - \text{Mode}}{\sigma}$$

When the mode is absent or unreliable, the median form is used.

$$Sk = \frac{3(\bar{x} - \text{Median})}{\sigma}$$

$$Sk = 0 \Rightarrow \text{Symmetric}$$

$$Sk > 0 \Rightarrow \text{Positively Skewed}$$

$$Sk < 0 \Rightarrow \text{Negatively Skewed}$$

Bowley's coefficient of skewness

Bowley's coefficient uses quartiles and is therefore less affected by extreme observations.

$$SkB = \frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1}$$

Central moments

Central moments describe the spread and shape of a distribution about its mean.

$$\mu_r = \frac{\sum (x_i - \bar{x})^r}{N}$$

$$\mu_r = \frac{\sum f_i (x_i - \bar{x})^r}{\sum f_i}$$

$$\mu_1 = \frac{\sum (x_i - \bar{x})}{N} = 0$$

$$\mu_2 = \frac{\sum (x_i - \bar{x})^2}{N} = \sigma^2$$

$$\mu_3 = \frac{\sum (x_i - \bar{x})^3}{N}$$

$$\mu_4 = \frac{\sum (x_i - \bar{x})^4}{N}$$

$$\sigma = \sqrt{\mu_2}$$

Skewness using moments

$$\gamma_1 = \frac{\mu_3}{\mu_2^{3/2}}$$

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3}$$

The sign of γ_1 or μ_3 gives the direction of skewness. Since μ_3 is squared in β_1 , β_1 is non-negative.

Kurtosis

Kurtosis describes the relative thickness of the tails of a distribution.

$$\beta_2 = \frac{\mu_4}{\mu_2^2}$$

$$\beta_2 = 3 \Rightarrow \text{Mesokurtic}$$

$$\beta_2 > 3 \Rightarrow \text{Leptokurtic}$$

$$\beta_2 < 3 \Rightarrow \text{Platykurtic}$$

Foundations of Probability

Random experiments, sample spaces, events, and probability rules

Random experiment and sample space

A random experiment has well-defined possible outcomes, but the exact outcome cannot be predicted in advance. The set of all possible outcomes is the sample space S . An event A is a subset of S .

$$S = \{\omega_1, \omega_2, \dots, \omega_n\}$$

$$A \subseteq S$$

Operations on events

$A \cup B$ represents A or B or both. $A \cap B$ represents A and B . A' or A^c represents not A .

$$A \cup B \quad A \cap B \quad A' = A^c$$

$$A \cap B = \emptyset \text{ (Mutually Exclusive)}$$

$$A \cup B = S \text{ (Collectively Exhaustive)}$$

$$A \cap A' = \emptyset \quad A \cup A' = S$$

Axioms of probability

$$P(A) \geq 0$$

$$P(S) = 1$$

$$0 \leq P(A) \leq 1$$

$$A \cap B = \emptyset \implies P(A \cup B) = P(A) + P(B)$$

Classical probability

When every possible outcome is equally likely, probability is the ratio of favourable outcomes to total outcomes.

$$P(A) = \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}}$$

$$P(A) = \frac{m}{n}$$

Complement rule

$$P(A') = 1 - P(A)$$

$$P(A) + P(A') = 1$$

$$P(\text{At least one}) = 1 - P(\text{None})$$

$$P(\text{At least one success}) = 1 - (1 - p)^n$$

Addition rule

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

The intersection is subtracted because it is counted once in $P(A)$ and again in $P(B)$. For mutually exclusive events, $P(A \cap B) = 0$.

$$A \cap B = \emptyset \implies P(A \cup B) = P(A) + P(B)$$

$$P(\text{Neither A nor B}) = 1 - P(A \cup B)$$

$$P(A \text{ only}) = P(A) - P(A \cap B)$$

Counting Techniques

Multiplication principle, permutations, combinations, and repeated objects

Fundamental principle of counting

If successive operations can be performed in $n_1, n_2, \dots, n_k$ ways, multiply the numbers of choices.

$$\text{Total Outcomes} = n_1 \times n_2 \times \cdots \times n_k$$

If restrictions create separate non-overlapping cases, calculate each case independently and then add the results.

Permutations

A permutation is an arrangement in which order matters.

$${}^n P_r = \frac{n!}{(n-r)!}$$

Permutations with identical objects

$$\text{Number of arrangements} = \frac{n!}{n_1! n_2! \cdots n_k!}$$

Combinations

A combination is a selection in which order does not matter.

$${}^n C_r = \frac{n!}{r!(n-r)!}$$

$${}^n P_r = {}^n C_r r!$$

Conditional Probability and Independence

Reduced sample spaces, product rules, and probability trees

Conditional probability

Conditional probability measures the probability of B after A is known to have occurred. The event after the vertical bar determines the reduced sample space.

$$P(B|A) = \frac{P(A \cap B)}{P(A)} \quad P(A) > 0$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad P(B) > 0$$

$$P(B|A) = \frac{n(A \cap B)}{n(A)}$$

Product rule

The product rule is used for intersections and sequential events. In a tree diagram, probabilities are multiplied along a branch.

$$P(A \cap B) = P(A)P(B|A)$$

$$P(A \cap B) = P(B)P(A|B)$$

$$P(A \cap B \cap C) = P(A)P(B|A)P(C|A \cap B)$$

Independent events

Events are independent when the occurrence of one does not change the probability of the other.

$$P(B|A) = P(B)$$

$$P(A|B) = P(A)$$

$$P(A \cap B) = P(A)P(B)$$

Independent events are not the same as mutually exclusive events. Without-replacement selections are normally dependent because the first selection changes the remaining sample space.

CHAPTER 9

Total Probability and Bayes' Theorem

Combining possible causes and reversing conditional probability

Partition of the sample space

Suppose $B_1, B_2, \dots, B_k$ are mutually exclusive and collectively exhaustive causes.

$$B_i \cap B_j = \emptyset, i \neq j$$

$$\bigcup_{i=1}^k B_i = S$$

Law of total probability

An outcome A may occur through several different causes. Multiply the probability of each cause by the conditional probability of A under that cause, then add all matching branches.

$$P(A) = \sum_{i=1}^k P(B_i)P(A|B_i)$$

$$P(A) = P(B_1)P(A|B_1) + P(B_2)P(A|B_2) + P(B_3)P(A|B_3)$$

Bayes' theorem

Bayes' theorem reverses the conditional direction. Once evidence A is observed, it calculates the probability that a particular cause B_r produced that evidence.

$$P(B_r|A) = \frac{P(B_r)P(A|B_r)}{P(A)}$$

$$P(B_r|A) = \frac{P(B_r)P(A|B_r)}{\sum_{i=1}^k P(B_i)P(A|B_i)}$$

$$P(B_r) = \text{Prior Probability}$$

$$P(A|B_r) = \text{Likelihood}$$

$$P(A) = \text{Evidence}$$

$$P(B_r|A) = \text{Posterior Probability}$$

Diagnostic-test form

Let C represent disease present and D represent a positive diagnosis. The total positive-diagnosis probability includes both true positives and false positives.

$$P(C \cap D) = P(C)P(D|C)$$

$$P(D) = P(C)P(D|C) + P(C')P(D|C')$$

$$P(C|D) = \frac{P(C)P(D|C)}{P(C)P(D|C) + P(C')P(D|C')}$$

Examination Method

A short procedure for selecting and applying the correct formula

Descriptive-statistics questions

First classify the data as ungrouped, discrete frequency, or continuous grouped. For continuous data calculate midpoints. Construct the required columns such as $f_i x_i$, cumulative frequency, deviations, squared deviations, or higher powers. Verify that $\sum f_i = N$ before using the formula.

Probability questions

Translate 'or' as $\cup$, 'and' as $\cap$, 'not' as a complement, and 'given' as conditional probability. Use the general addition rule unless the events are definitely mutually exclusive. For a probability tree, multiply along each branch and add branches that lead to the required outcome.

Interpretation

End the calculation with its meaning. State the units of a statistical measure, the direction of skewness, the type of kurtosis, which dataset is more consistent, or the meaning of a conditional or posterior probability.

Worksheet 1 - Fully Worked Solutions

Measures of central tendency and positional measures

W1.1 Arithmetic mean - company orders

Question. The number of new orders received by a company over the last 25 working days were recorded as follows: 3, 0, 1, 4, 4, 4, 2, 5, 3, 6, 4, 5, 1, 4, 2, 3, 0, 2, 0, 5, 4, 2, 3, 3, 1. Calculate the arithmetic mean for the number of orders received over all similar working days.

$$N = 25 \quad \Sigma x_i = 71$$

$$\bar{x} = \frac{\Sigma x_i}{N}$$

$$\bar{x} = \frac{71}{25} = 2.84$$

Final answer. The arithmetic mean is 2.84 orders per working day.

W1.2 Arithmetic mean - nurses' overtime

Question. The human resource manager at a city hospital began a study of the overtime hours of the registered nurses. Fifteen nurses were selected at random, and the following overtime hours during a month were recorded: 13, 13, 12, 15, 7, 15, 5, 12, 6, 7, 12, 10, 9, 13, 12, 5, 9, 6, 10, 5, 6, 9, 6, 9, 12. Compute the arithmetic mean of overtime hours during the month.

Although the question says fifteen nurses, 25 observations are printed. The calculation therefore uses all 25 listed observations.

$$N = 25 \quad \Sigma x_i = 238$$

$$\bar{x} = \frac{238}{25} = 9.52$$

Final answer. The arithmetic mean is 9.52 hours.

W1.3 Mean - continuous accident data

Question. A company is planning to improve plant safety. For this, accident data for the last 50 weeks was compiled. These data are grouped into the frequency distribution shown below. Calculate the A.M. of the number of accidents per week. Number of accidents: 0–4, 5–9, 10–14, 15–19, 20–24; Number of weeks: 5, 22, 13, 8, 2.

$$x_i = 2, 7, 12, 17, 22$$

$$\Sigma f_i = 50 \quad \Sigma f_i x_i = 5(2) + 22(7) + 13(12) + 8(17) + 2(22) = 500$$

$$\bar{x} = \frac{\Sigma f_i x_i}{\Sigma f_i}$$

$$\bar{x} = \frac{500}{50} = 10$$

Final answer. The mean is 10 accidents per week.

W1.4 Mean by step deviation - overtime hours

Question. The following distribution gives the pattern of overtime work done by 100 employees of a company. Calculate the average overtime work done per employee (using shortcut/step-deviation method). Overtime hours: 10–15, 15–20, 20–25, 25–30, 30–35, 35–40; Number of employees: 11, 20, 35, 20, 8, 6.

$$x_i = 12.5, 17.5, 22.5, 27.5, 32.5, 37.5$$

$$A = 22.5 \quad h = 5 \quad u_i = -2, -1, 0, 1, 2, 3$$

$$\Sigma f_i u_i = 11(-2) + 20(-1) + 35(0) + 20(1) + 8(2) + 6(3) = 12$$

$$\bar{x} = A + \frac{\Sigma f_i u_i}{\Sigma f_i} \times h$$

$$\bar{x} = 22.5 + \frac{12}{100} \times 5 = 23.1$$

Final answer. The average overtime is 23.1 hours.

W1.5 Correcting a mean - one error

Question. The average dividend declared by a group of 10 chemical companies was 18 per cent. Later on, it was discovered that one correct figure, 12, was misread as 22. Find the correct average dividend.

$$\Sigma x_{\text{wrong}} = 10(18) = 180$$

$$\Sigma x_{\text{correct}} = 180 - 22 + 12 = 170$$

$$\bar{x}_{\text{correct}} = \frac{170}{10} = 17\%$$

Final answer. The corrected average dividend is 17%.

W1.6 Correcting a mean - two errors

Question. The mean of 200 observations was 50. Later on, it was found that two observations were misread as 92 and 8 instead of 192 and 88. Find the correct mean.

$$\Sigma x_{\text{wrong}} = 200(50) = 10,000$$

$$\Sigma x_{\text{correct}} = 10,000 - 92 - 8 + 192 + 88 = 10,180$$

$$\bar{x}_{\text{correct}} = \frac{10,180}{200} = 50.9$$

Final answer. The corrected mean is 50.9.

W1.7 Median - ungrouped data

Question. Calculate the median of the following data that relates to the number of patients examined per hour in the outpatient ward (OPD) in a hospital: 10, 12, 15, 20, 13, 24, 17, 18.

$$10, 12, 13, 15, 17, 18, 20, 24$$

$$N = 8$$

$$\text{Median} = \frac{x_4 + x_5}{2}$$

$$\text{Median} = \frac{15 + 17}{2} = 16$$

Final answer. The median is 16 patients per hour.

W1.8 Median wage - grouped data

Question. In a factory employing 3000 persons, 5 per cent earn less than Rs 150 per day, 580 earn from Rs 151 to Rs 200 per day, 30 per cent earn from Rs 201 to Rs 250 per day, 500 earn from Rs 251 to Rs 300 per day, 20 per cent earn from Rs 301 to Rs 350 per day, and the rest earn Rs 351 or more per day. What is the median wage?

$$f = 150, 580, 900, 500, 600, 270$$

$$CF = 150, 730, 1630, 2130, 2730, 3000$$

$$N/2 = 1500 \text{ Median class} = 201-250$$

$$L = 200.5 \text{ } CF_{<} = 730 \text{ } f_m = 900 \text{ } h = 50$$

$$\text{Median} = 200.5 + \frac{1500 - 730}{900} \times 50$$

$$\text{Median} = 200.5 + 42.7778 = 243.2778$$

Final answer. The median wage is Rs 243.28 per day.

W1.9 Quartiles and P75 - graphical distribution

Question. You are working for the transport manager of a 'call centre' which hires cars for the staff. You are interested in the weekly distances covered by these cars. Kilometers recorded for a sample of hired cars during a given week yielded the following data. Kilometers covered: 100–110, 110–120, 120–130, 130–140, 140–150, 150–160, 160–170, 170–180, 180–190; Number of cars: 4, 0, 3, 7, 11, 8, 5, 0, 2. (a) Form a cumulative frequency distribution. (b) Calculate Q_1 , Q_2 , Q_3 and P_{75} .

$$CF = 4, 4, 7, 14, 25, 33, 38, 38, 40 \text{ } N = 40$$

$$Q_1 = 130 + \frac{10 - 7}{7} \times 10 = 134.29$$

$$Q_2 = 140 + \frac{20 - 14}{11} \times 10 = 145.45$$

$$Q_3 = 150 + \frac{30 - 25}{8} \times 10 = 156.25$$

$$P_{75} = Q_3 = 156.25$$

Answer-key check. Printed answer: $Q_1 = 135$ km, $Q_2 = 145$ km, $Q_3 = P_{75} = 155$ km (graphical readings)
Verified from the supplied data: Formula interpolation gives $Q_1 = 134.29$ km, $Q_2 = 145.45$ km, $Q_3 = P_{75} = 156.25$ km. A hand-drawn ogive may produce the printed approximations.

W1.10 Q1, D7, and P60 - grouped overtime

Question. The following distribution gives the pattern of overtime work per week done by 100 employees of a company. Calculate Q_1 , D_7 , and P_{60} . Overtime hours: 10–15, 15–20, 20–25, 25–30, 30–35, 35–40; No. of employees: 11, 20, 35, 20, 8, 6.

$$CF = 11, 31, 66, 86, 94, 100 \quad N = 100$$

$$Q_1 = 15 + \frac{25 - 11}{20} \times 5 = 18.5$$

$$D_7 = 25 + \frac{70 - 66}{20} \times 5 = 26$$

$$P_{60} = 20 + \frac{60 - 31}{35} \times 5 = 24.14$$

Answer-key check. Printed answer: $Q_1 = 18.5$, $D_7 = 25.57$, $P_{60} = 24.14$

Verified from the supplied data: $Q_1 = 18.5$ hours, $D_7 = 26.00$ hours, $P_{60} = 24.14$ hours. The printed D_7 does not follow the supplied frequency table.

Worksheet 2 - Fully Worked Solutions

Dispersion, skewness, kurtosis, and moments

W2.1 Range - monthly sales

Question. The following are the sales figures of a firm for the last 12 months. Months: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12; Sales (Rs '000): 80, 82, 82, 84, 84, 86, 86, 88, 88, 90, 90, 92. Calculate the range and coefficient of range for sales.

$$R = X_{\max} - X_{\min} = 92 - 80 = 12$$

$$\text{Coefficient of Range} = \frac{92 - 80}{92 + 80} = 0.0698$$

Final answer. Range = 12; coefficient of range = 0.0698.

W2.2 Range - daily wages

Question. The following data represents the daily wages (in Rs) of 10 workers: 120, 125, 130, 135, 140, 145, 150, 155, 160, 165. Calculate the range and coefficient of range.

$$R = 165 - 120 = 45$$

$$\text{Coefficient of Range} = \frac{165 - 120}{165 + 120} = 0.1579$$

Final answer. Range = Rs 45; coefficient of range = 0.1579.

W2.3 Variance and SD - commodity prices

Question. The wholesale prices of a commodity for seven consecutive days in a month is as follows. Days: 1, 2, 3, 4, 5, 6, 7; Commodity price/unit: 240, 260, 270, 245, 255, 286, 264. Calculate the variance and standard deviation.

$$N = 7 \quad \Sigma X_i = 1820 \quad \bar{x} = 260$$

$$\Sigma (x_i - \bar{x})^2 = 1442$$

$$\sigma^2 = \frac{1442}{7} = 206$$

$$\sigma = \sqrt{206} = 14.35$$

Answer-key check. Printed answer: Variance = 196.49; SD = 14.02

Verified from the supplied data: Variance = 206.00; SD = 14.35 from the seven printed observations.

W2.4 Standard deviation - student marks

Question. Calculate the standard deviation for the following marks obtained by 10 students: 12, 15, 18, 20, 22, 25, 28, 30, 32, 35.

$$N = 10 \quad \Sigma x_i = 237 \quad \bar{x} = 23.7$$

$$\Sigma (x_i - \bar{x})^2 = 518.1$$

$$\sigma^2 = \frac{518.1}{10} = 51.81$$

$$\sigma = \sqrt{51.81} = 7.20$$

Final answer. The population standard deviation is 7.20 marks (the worksheet rounds it to 7.18).

W2.5 Standard deviation - engineering-company profits

Question. A study of 100 engineering companies gives the following information. Profit (Rs in crore): 0–10, 10–20, 20–30, 30–40, 40–50, 50–60; Number of companies: 8, 12, 20, 30, 20, 10. Calculate the standard deviation of the profit earned.

$$x_i = 5, 15, 25, 35, 45, 55 \quad N = 100$$

$$\Sigma f_i x_i = 3220 \quad \bar{x} = 32.2$$

$$\Sigma f_i (x_i - \bar{x})^2 = 19,216$$

$$\sigma^2 = \frac{19,216}{100} = 192.16$$

$$\sigma = \sqrt{192.16} = 13.86$$

Answer-key check. Printed answer: SD = 14.91 crore

Verified from the supplied data: SD = 13.86 crore from the printed classes and frequencies.

W2.6 Variance and SD - weekly accidents

Question. The following distribution shows the number of accidents per week in a factory. Number of accidents: 0–4, 5–9, 10–14, 15–19, 20–24; Number of weeks: 5, 22, 13, 8, 2. Calculate the variance and standard deviation.

$$x_i = 2, 7, 12, 17, 22 \quad \bar{x} = 10$$

$$\sum f_i (x_i - \bar{x})^2 = 1250$$

$$\sigma^2 = \frac{1250}{50} = 25$$

$$\sigma = \sqrt{25} = 5$$

Answer-key check. Printed answer: Variance = 24.0; SD = 4.90

Verified from the supplied data: Variance = 25.00; SD = 5.00 from the printed distribution.

W2.7 Coefficient of variation - products A and B

Question. The weekly sales of two products A and B were recorded as given below. Product A: 59, 75, 27, 63, 27, 28, 56; Product B: 150, 200, 125, 310, 330, 250, 225. Find out which of the two shows greater fluctuation in sales.

$$\bar{x}_a = 47.857 \quad \sigma_a = 18.597$$

$$CV_a = \frac{18.597}{47.857} \times 100 = 38.86\%$$

$$\bar{x}_\beta = 227.143 \quad \sigma_\beta = 70.804$$

$$CV_\beta = \frac{70.804}{227.143} \times 100 = 31.17\%$$

$$CV_a > CV_\beta$$

Final answer. Product A has greater relative fluctuation. The printed CV values differ, but the printed conclusion is correct.

W2.8 Consistency - two factories

Question. The mean and standard deviation of two factories are given below. Factory A: mean = 500, standard deviation = 40; Factory B: mean = 600, standard deviation = 45. Which factory has more consistent performance?

$$CV_a = \frac{40}{500} \times 100 = 8\%$$

$$CV_\beta = \frac{45}{600} \times 100 = 7.5\%$$

$$CV_\beta < CV_a$$

Answer-key check. Printed answer: Factory A is more consistent
 Verified from the supplied data: Factory B is more consistent because $7.5\% < 8\%$.

W2.9 Mean deviation - quantity demanded

Question. Find the average deviation from mean for the following distribution. Quantity demanded (in units): 60, 61, 62, 63, 64, 65, 66, 67, 68; Frequency: 2, 0, 15, 29, 25, 12, 10, 4, 3.

$$N = 100 \quad \Sigma f_i x_i = 6389 \quad \bar{x} = 63.89$$

$$\Sigma f_i |x_i - \bar{x}| = 123.88$$

$$MD\bar{x} = \frac{123.88}{100} = 1.2388$$

Answer-key check. Printed answer: MD = 1.19 units
 Verified from the supplied data: MD = 1.24 units from the printed distribution.

W2.10 Mean deviation - discrete grouped data

Question. Calculate the mean deviation from mean for the following data. x: 10, 20, 30, 40, 50; f: 5, 8, 12, 7, 3.

$$N = 35 \quad \Sigma f_i x_i = 1000 \quad \bar{x} = 28.5714$$

$$\Sigma f_i |x_i - \bar{x}| = 322.8571$$

$$MD\bar{x} = \frac{322.8571}{35} = 9.2245$$

Answer-key check. Printed answer: MD = 9.71
 Verified from the supplied data: MD = 9.22 from the printed distribution.

W2.11 Karl Pearson skewness - rejected items

Question. Data of rejected items during a production process is as follows. No. of rejects (per operator): 21–25, 26–30, 31–35, 36–40, 41–45, 46–50, 51–55; No. of operators: 5, 15, 28, 42, 15, 12, 3. Calculate the mean, standard deviation, and coefficient of skewness and comment on the results.

$$x_i = 23, 28, 33, 38, 43, 48, 53 \quad N = 120$$

$$\Sigma f_i x_i = 4435 \quad \bar{x} = 36.9583$$

$$\sigma^2 = 45.3733 \quad \sigma = 6.7360$$

$$\text{Mode} = 35.5 + \frac{42 - 28}{2(42) - 28 - 15} \times 5 = 37.2073$$

$$Sk = \frac{36.9583 - 37.2073}{6.7360} = -0.037$$

Final answer. Mean = 36.96, SD = 6.74, Sk = -0.04; the distribution is approximately symmetric.

W2.12 Karl Pearson skewness - symmetric distribution

Question. From the following data, calculate Karl Pearson's coefficient of skewness. Class: 0–10, 10–20, 20–30, 30–40, 40–50; Frequency: 5, 10, 20, 10, 5.

$$\bar{x} = 25 \quad \text{Mode} = 25$$

$$Sk = \frac{25 - 25}{\sigma} = 0$$

Final answer. The coefficient of skewness is 0; the distribution is perfectly symmetric.

W2.13(a) CV and skewness from summary statistics

Question. The sum of 50 observations is 500, its sum of squares is 6000 and median 12. Find the coefficient of variation and coefficient of skewness.

$$\bar{x} = \frac{500}{50} = 10$$

$$\sigma^2 = 6000/50 - 10^2 = 20 \quad \sigma = \sqrt{20} = 4.4721$$

$$CV = \frac{4.4721}{10} \times 100 = 44.72\%$$

$$Sk = \frac{3(10 - 12)}{4.4721} = -1.342$$

Answer-key check. Printed answer: CV = 20%; Sk = 0

Verified from the supplied data: CV = 44.72%; Sk = -1.342 from the supplied summary statistics.

W2.13(b) Mode and median from CV and skewness

Question. For a moderately skewed distribution, the arithmetic mean is 100 and coefficient of variation 35, and Pearson's coefficient of skewness is 0.2. Find the mode and the median.

$$35 = (\sigma/100)100 \quad \sigma = 35$$

$$0.2 = (100 - \text{Mode})/35 \quad \text{Mode} = 93$$

$$0.2 = 3(100 - \text{Median})/35 \quad \text{Median} = 97.67$$

Final answer. Mode = 93; Median = 97.67 (approximately 98). The worksheet prints 97.

W2.14 Skewness and kurtosis from moments

Question. The first four central moments of a distribution are 0, 16, -10, and 240 respectively. Calculate the coefficient of skewness and kurtosis.

$$\mu_1 = 0 \quad \mu_2 = 16 \quad \mu_3 = -10 \quad \mu_4 = 240$$

$$\gamma_1 = \frac{-10}{16^{3/2}} = -0.15625$$

$$\beta_1 = \frac{(-10)^2}{16^3} = 0.02441$$

$$\beta_2 = \frac{240}{16^2} = 0.9375$$

Answer-key check. Printed answer: $\beta_1 = -0.0977$; $\beta_2 = 0.9375$

Verified from the supplied data: Signed skewness $\gamma_1 = -0.15625$; $\beta_1 = 0.02441$; $\beta_2 = 0.9375$.

W2.15 SD and kurtosis - electricity consumption

Question. Find the standard deviation and kurtosis of the following set of data pertaining to kilowatt hours (kWh) of electricity consumed by 100 persons in a city. Consumption (in kWh): 0–10, 10–20, 20–30, 30–40, 40–50; Number of users: 10, 20, 40, 20, 10.

$$\bar{x} = 25 \quad \mu_2 = 120 \quad \mu_4 = 36,000$$

$$\sigma = \sqrt{120} = 10.95$$

$$\beta_2 = \frac{36,000}{120^2} = 2.50$$

$$\beta_2 < 3 \Rightarrow \text{Platykurtic}$$

Answer-key check. Printed answer: SD = 12.25; $\beta_2 = 2.74$

Verified from the supplied data: SD = 10.95 kWh; $\beta_2 = 2.50$ from the supplied table.

W2.16 Kurtosis - continuous data

Question. Calculate the coefficient of kurtosis for the following data. Class: 10–20, 20–30, 30–40, 40–50; Frequency: 6, 12, 18, 4.

$$\bar{x} = 30 \quad \mu_2 = 75 \quad \mu_4 = 13,125$$

$$\beta_2 = \frac{13,125}{75^2} = 2.3333$$

Final answer. $\beta_2 = 2.33$; therefore the distribution is platykurtic.

W2.17 Moments - continuous frequency distribution

Question. Find the variance, skewness, and kurtosis of the following frequency distribution by the method of moments. Class interval: 0–10, 10–20, 20–30, 30–40; Frequency: 1, 4, 3, 2.

$$x_i = 5, 15, 25, 35 \quad N = 10 \quad \bar{x} = 21$$

$$\mu_2 = 84 \quad \mu_3 = 72 \quad \mu_4 = 14,832$$

$$\gamma_1 = \frac{72}{84^{3/2}} = 0.09352$$

$$\beta_1 = \frac{72^2}{84^3} = 0.00875$$

$$\beta_2 = \frac{14,832}{84^2} = 2.1020$$

Answer-key check. Printed answer: $\mu_2 = 81.6$; $\beta_1 = 0.078$; $\beta_2 = 1.81$

Verified from the supplied data: $\mu_2 = 84$; $\gamma_1 = 0.09352$; $\beta_1 = 0.00875$; $\beta_2 = 2.1020$ from the printed distribution.

W2.18 Coefficients from central moments

Question. For a frequency distribution, the first four moments about mean are 0, 4, 6, and 40 respectively. Calculate the coefficient of skewness and kurtosis.

$$\mu_1 = 0 \quad \mu_2 = 4 \quad \mu_3 = 6 \quad \mu_4 = 40$$

$$\gamma_1 = \frac{6}{4^{3/2}} = 0.75$$

$$\beta_1 = \frac{6^2}{4^3} = 0.5625$$

$$\beta_2 = \frac{40}{4^2} = 2.5$$

Final answer. $\beta_1 = 0.5625$; $\beta_2 = 2.5$. The signed skewness is $\gamma_1 = 0.75$.

Worksheet 3 - Fully Worked Solutions

Probability, conditional probability, total probability, and Bayes' theorem

W3.1 Inclusion-exclusion - Shanghai and Beijing

Question. The probability that an American industry will locate in Shanghai, China, is 0.7, the probability that it will locate in Beijing, China, is 0.4, and the probability that it will locate in either Shanghai or Beijing or both is 0.8. What is the probability that the industry will locate (a) in both cities? (b) in neither city?

$$P(S \cap B) = P(S) + P(B) - P(S \cup B)$$

$$P(S \cap B) = 0.7 + 0.4 - 0.8 = 0.3$$

$$P(\text{Neither}) = 1 - P(S \cup B) = 1 - 0.8 = 0.2$$

Final answer. $P(\text{Both}) = 0.3$; $P(\text{Neither}) = 0.2$.

W3.2 Inclusion-exclusion - bonds and mutual funds

Question. From past experience, a stockbroker believes that under present economic conditions a customer will invest in tax-free bonds with a probability of 0.6, will invest in mutual funds with a probability of 0.3, and will invest in both tax-free bonds and mutual funds with a probability of 0.15. At this time, find the probability that a customer will invest (a) in either tax-free bonds or mutual funds; (b) in neither tax-free bonds nor mutual funds.

$$P(T \cup M) = 0.6 + 0.3 - 0.15 = 0.75$$

$$P(\text{Neither}) = 1 - 0.75 = 0.25$$

Final answer. $P(\text{Either}) = 0.75$; $P(\text{Neither}) = 0.25$.

W3.3 Conditional probability - education and gender

Question. A random sample of 200 adults are classified below by sex and their level of education attained. Male: Elementary 38, Secondary 28, College 22; Female: Elementary 45, Secondary 50, College 17. If a person is picked at random from this group, find the probability that (a) the person is a male, given that the person has a secondary education; (b) the person does not have a college degree, given that the person is a female.

$$P(\text{Male}|\text{Secondary}) = \frac{28}{28 + 50} = 14/39 = 0.359$$

$$P(\text{No College}|\text{Female}) = \frac{45 + 50}{45 + 50 + 17} = 95/112 = 0.848$$

Final answer. $14/39 \approx 0.359$ and $95/112 \approx 0.848$.

W3.4 Conditional probability - hypertension and smoking

Question. In an experiment to study the relationship of hypertension and smoking habits, the following data are collected for 180 individuals. Hypertension: Nonsmokers 21, Moderate Smokers 36, Heavy Smokers 30; Nonhypertension: Nonsmokers 48, Moderate Smokers 26, Heavy Smokers 19. If one of these individuals is selected at random, find the probability that the person is (a) experiencing hypertension, given that the person is a heavy smoker; (b) a nonsmoker, given that the person is experiencing no hypertension.

$$P(\text{Hyp}|\text{Heavy}) = \frac{30}{30 + 19} = 30/49 = 0.612$$

$$P(\text{Nonsmoker}|\text{NH}) = \frac{48}{48 + 26 + 19} = 16/31 = 0.516$$

Final answer. $30/49 \approx 0.612$ and $16/31 \approx 0.516$.

W3.5 Conditional probability - oil and filter

Question. The probability that an automobile being filled with gasoline also needs an oil change is 0.25; the probability that it needs a new oil filter is 0.40; and the probability that both the oil and the filter need changing is 0.14. (a) If the oil has to be changed, what is the probability that a new oil filter is needed? (b) If a new oil filter is needed, what is the probability that the oil has to be changed?

$$P(F|O) = \frac{P(O \cap F)}{P(O)} = 0.14/0.25 = 0.56$$

$$P(O|F) = \frac{P(O \cap F)}{P(F)} = 0.14/0.40 = 0.35$$

Final answer. $P(F|O) = 0.56$; $P(O|F) = 0.35$.

W3.6 Conditional probability - television show

Question. The probability that a married man watches a certain television show is 0.4, and the probability that a married woman watches the show is 0.5. The probability that a man watches the show, given that his wife does, is 0.7. Find the probability that (a) a married couple watches the show; (b) a wife watches the show, given that her husband does; (c) at least one member of a married couple will watch the show.

$$P(M \cap W) = P(W)P(M|W) = 0.5(0.7) = 0.35$$

$$P(W|M) = \frac{P(M \cap W)}{P(M)} = 0.35/0.4 = 0.875$$

$$P(M \cup W) = 0.4 + 0.5 - 0.35 = 0.55$$

Final answer. Both = 0.35; $P(W|M) = 0.875$; at least one = 0.55.

W3.7 Conditional probability - bond referendum

Question. For married couples living in a certain suburb, the probability that the husband will vote on a bond referendum is 0.21, the probability that the wife will vote on the referendum is 0.28, and the probability that both the husband and the wife will vote is 0.15. What is the probability that (a) at least one member of a married couple will vote? (b) a wife will vote, given that her husband will vote? (c) a husband will vote, given that his wife will not vote?

$$P(H \cup W) = 0.21 + 0.28 - 0.15 = 0.34$$

$$P(W|H) = \frac{0.15}{0.21} = 5/7$$

$$P(H \cap W') = P(H) - P(H \cap W) = 0.06$$

$$P(H|W') = \frac{0.06}{1 - 0.28} = 1/12$$

Final answer. 0.34, $5/7 \approx 0.714$, and $1/12 \approx 0.0833$.

W3.8 Total probability - cancer diagnosis

Question. In a certain region of the country it is known from past experience that the probability of selecting an adult over 40 years of age with cancer is 0.05. If the probability of a doctor correctly diagnosing a person with cancer as having the disease is 0.78 and the probability of incorrectly diagnosing a person without cancer as having the disease is 0.06, what is the probability that an adult over 40 years of age is diagnosed as having cancer?

$$P(C') = 1 - 0.05 = 0.95$$

$$P(D) = P(C)P(D|C) + P(C')P(D|C')$$

$$P(D) = 0.05(0.78) + 0.95(0.06) = 0.039 + 0.057 = 0.096$$

Final answer. $P(D) = 0.096$.

W3.9 Bayes' theorem - cancer after diagnosis

Question. Referring to the previous problem, what is the probability that a person diagnosed as having cancer actually has the disease?

$$P(C|D) = \frac{P(C)P(D|C)}{P(D)}$$

$$P(C|D) = \frac{0.05(0.78)}{0.096} = 0.40625$$

Final answer. $P(C|D) = 0.40625 = 40.625\%$.

W3.10 Bayes' theorem - unstamped film packages

Question. Suppose that the four inspectors at a film factory are supposed to stamp the expiration date on each package of film at the end of the assembly line. John, who stamps 20% of the packages, fails to stamp the expiration date once in every 200 packages; Tom, who stamps 60% of the packages, fails to stamp the expiration date once in every 100 packages; Jeff, who stamps 15% of the packages, fails to stamp the expiration date once in every 90 packages; and Pat, who stamps 5% of the packages, fails to stamp the expiration date once in every 200 packages. If a customer complains that her package of film does not show the expiration date, what is the probability that it was inspected by John?

$$P(F) = 0.20(1/200) + 0.60(1/100) + 0.15(1/90) + 0.05(1/200)$$

$$P(F) = 107/12,000$$

$$P(J|F) = \frac{0.20(1/200)}{107/12,000} = 12/107 = 0.11215$$

Answer-key check. Printed answer: $20/143 \approx 0.1399$

Verified from the supplied data: $12/107 \approx 0.11215$ from the stated workloads and failure rates.

W3.11 Conditional probability - relay stations

Question. A regional telephone company operates three identical relay stations at different locations. During a one-year period, the number of malfunctions reported by each station and the causes are shown below. Problems with electricity supplied: A 2, B 1, C 1; Computer malfunction: A 4, B 3, C 2; Malfunctioning electrical equipment: A 5, B 4, C 2; Caused by other human errors: A 7, B 7, C 5. Suppose that a malfunction was reported and it was found to be caused by other human errors. What is the probability that it came from station C?

$$P(C|\text{Human Error}) = \frac{5}{7 + 7 + 5} = 5/19 = 0.2632$$

Final answer. The probability is $5/19 \approx 0.263$.

W3.12 Bayes' theorem - latex paint and rollers

Question. A paint-store chain produces and sells latex and semigloss paint. Based on long-range sales, the probability that a customer will purchase latex paint is 0.75. Of those that purchase latex paint, 60% also purchase rollers. But only 30% of semigloss paint buyers purchase rollers. A randomly selected buyer purchases a roller and a can of paint. What is the probability that the paint is latex?

$$P(R) = 0.75(0.60) + 0.25(0.30) = 0.525$$

$$P(L|R) = \frac{0.75(0.60)}{0.525} = 6/7 = 0.8571$$

Final answer. The probability is $6/7 \approx 0.857$.

W3.13 Monty Hall

Question. Denote by A, B, and C the events that a grand prize is behind doors A, B, and C, respectively. Suppose you randomly picked a door, say A. The game host opened a door, say B, and showed there was no prize behind it. Now the host offers you the option of either staying at the door that you picked (A) or switching to the remaining unopened door (C). Use probability to explain whether you should switch or not.

$$P(\text{Prize behind A}) = 1/3$$

$$P(\text{Prize not behind A}) = 1 - 1/3 = 2/3$$

$$P(\text{Win by staying}) = 1/3$$

$$P(\text{Win by switching}) = 2/3$$

Final answer. Switch doors. The probability of winning by switching is $2/3$.

W3.14 Bayes' theorem - truth serum

Question. A truth serum has the property that 90% of the guilty suspects are properly judged while, of course, 10% of the guilty suspects are improperly found innocent. On the other hand, innocent suspects are misjudged 1% of the time. If the suspect was selected from a group of suspects of which only 5% have ever committed a crime, and the serum indicates that he is guilty, what is the probability that he is innocent?

$$P(I) = 0.95 \quad P(+|G) = 0.90 \quad P(+|I) = 0.01$$

$$P(+) = 0.05(0.90) + 0.95(0.01) = 0.0545$$

$$P(I|+) = \frac{0.95(0.01)}{0.0545} = 19/109 = 0.17431$$

Answer-key check. Printed answer: $0.155 = 15.5\%$

Verified from the supplied data: $19/109 \approx 0.17431 = 17.43\%$ from the stated probabilities.

W3.15 Bayes' theorem - faulty motel plumbing

Question. A large industrial firm uses three local motels to provide overnight accommodations for its clients. From past experience it is known that 20% of the clients are assigned rooms at the Ramada Inn, 50% at the Sheraton, and 30% at the Lakeview Motor Lodge. If the plumbing is faulty in 5% of the rooms at the Ramada Inn, in 4% of the rooms at the Sheraton, and in 8% of the rooms at the Lakeview Motor Lodge, what is the probability that (a) a client will be assigned a room with faulty plumbing? (b) a person with a room having faulty plumbing was assigned accommodations at the Lakeview Motor Lodge?

$$P(F) = 0.20(0.05) + 0.50(0.04) + 0.30(0.08) = 0.054$$

$$P(L|F) = \frac{0.30(0.08)}{0.054} = 4/9 = 0.4444$$

Final answer. $P(F) = 0.054$; $P(L|F) = 4/9 \approx 0.444$.

W3.16 Bayes' theorem - consulting-firm overruns

Question. A certain federal agency employs three consulting firms (A, B, and C) with probabilities 0.40, 0.35, and 0.25, respectively. From past experience it is known that the probability of cost overruns for the firms are 0.05, 0.03, and 0.15, respectively. Suppose a cost overrun is experienced by the agency. (a) What is the probability that the consulting firm involved is company C? (b) What is the probability that it is company A?

$$P(O) = 0.40(0.05) + 0.35(0.03) + 0.25(0.15) = 0.068$$

$$P(C|O) = \frac{0.25(0.15)}{0.068} = 75/136 = 0.55147$$

$$P(A|O) = \frac{0.40(0.05)}{0.068} = 5/17 = 0.29412$$

Answer-key check. Printed answer: $P(C|O) = 75/134$; $P(A|O) = 10/67$
 Verified from the supplied data: $P(C|O) = 75/136 \approx 0.5515$; $P(A|O) = 5/17 \approx 0.2941$.

W3.17 Bayes' theorem - disease after a negative test

Question. A certain form of cancer is known to be found in women over 60 with probability 0.07. A blood test exists for the detection of the disease, but the test is not infallible. In fact, it is known that 10% of the time the test gives a false negative (i.e., the test incorrectly gives a negative result) and 5% of the time the test gives a false positive (i.e., incorrectly gives a positive result). If a woman over 60 is known to have taken the test and received a favorable (i.e., negative) result, what is the probability that she has the disease?

$$P(-|C) = 0.10 \quad P(-|C') = 0.95$$

$$P(-) = 0.07(0.10) + 0.93(0.95) = 0.8905$$

$$P(C|-) = \frac{0.07(0.10)}{0.8905} = 14/1781 = 0.00786$$

Final answer. The probability is $0.00786 \approx 0.786\%$.

W3.18 Bayes' theorem - defectives in lots

Question. A producer of a certain type of electronic component ships to suppliers in lots of twenty. Suppose that 60% of all such lots contain no defective components, 30% contain one defective component, and 10% contain two defective components. A lot is picked, two components from the lot are randomly selected and tested, and neither is defective. (a) What is the probability that zero defective components exist in the lot? (b) What is the probability that one defective exists in the lot? (c) What is the probability that two defectives exist in the lot?

$$P(NN|D_0) = 1$$

$$P(NN|D_1) = \frac{C(19,2)}{C(20,2)} = 171/190 = 0.9$$

$$P(NN|D_2) = \frac{C(18,2)}{C(20,2)} = 153/190$$

$$P(NN) = 0.60(1) + 0.30(0.9) + 0.10(153/190) = 903/950$$

$$P(D_0|NN) = \frac{0.60(1)}{903/950} = 190/301 = 0.63123$$

$$P(D_1|NN) = \frac{0.30(0.9)}{903/950} = 171/602 = 0.28405$$

$$P(D_2|NN) = \frac{0.10(153/190)}{903/950} = 51/602 = 0.08472$$

Answer-key check. Printed answer: 285/313, 28/313, 0

Verified from the supplied data: $190/301 \approx 0.63123$, $171/602 \approx 0.28405$, and $51/602 \approx 0.08472$.

W3.19 Bayes' theorem - rare disease

Question. A rare disease exists with which only 1 in 500 people is affected. A test for the disease exists, but it is not infallible. A correct positive result (patient actually has the disease) occurs 95% of the time, while a false positive result (patient does not have the disease) occurs 5% of the time. If a randomly selected person takes the test and tests positive, what is the probability that they actually have the disease?

$$P(C) = 1/500 = 0.002 \quad P(C') = 0.998$$

$$P(+) = 0.002(0.95) + 0.998(0.05) = 0.0518$$

$$P(C|+) = \frac{0.002(0.95)}{0.0518} = 19/518 = 0.03668$$

Final answer. The probability is $0.03668 \approx 3.67\%$.